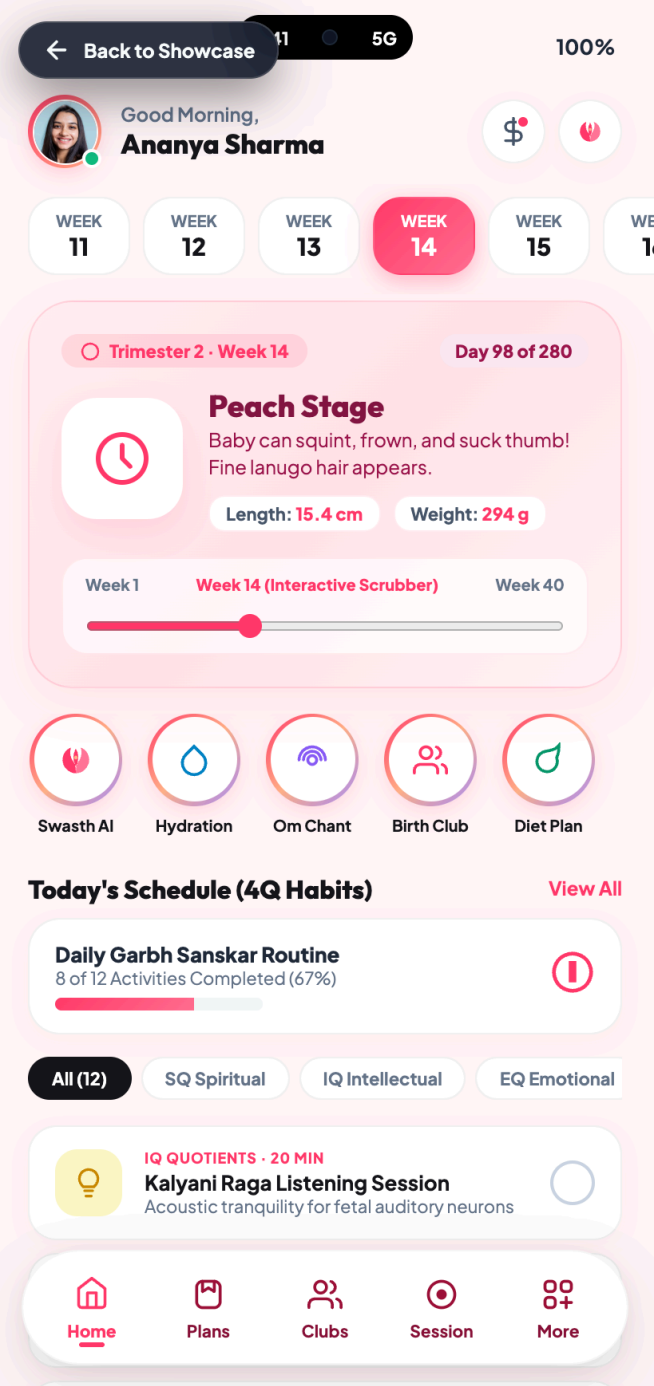


Matru-Kamal SWASTH HACKATHON 2026 · APP DESIGN

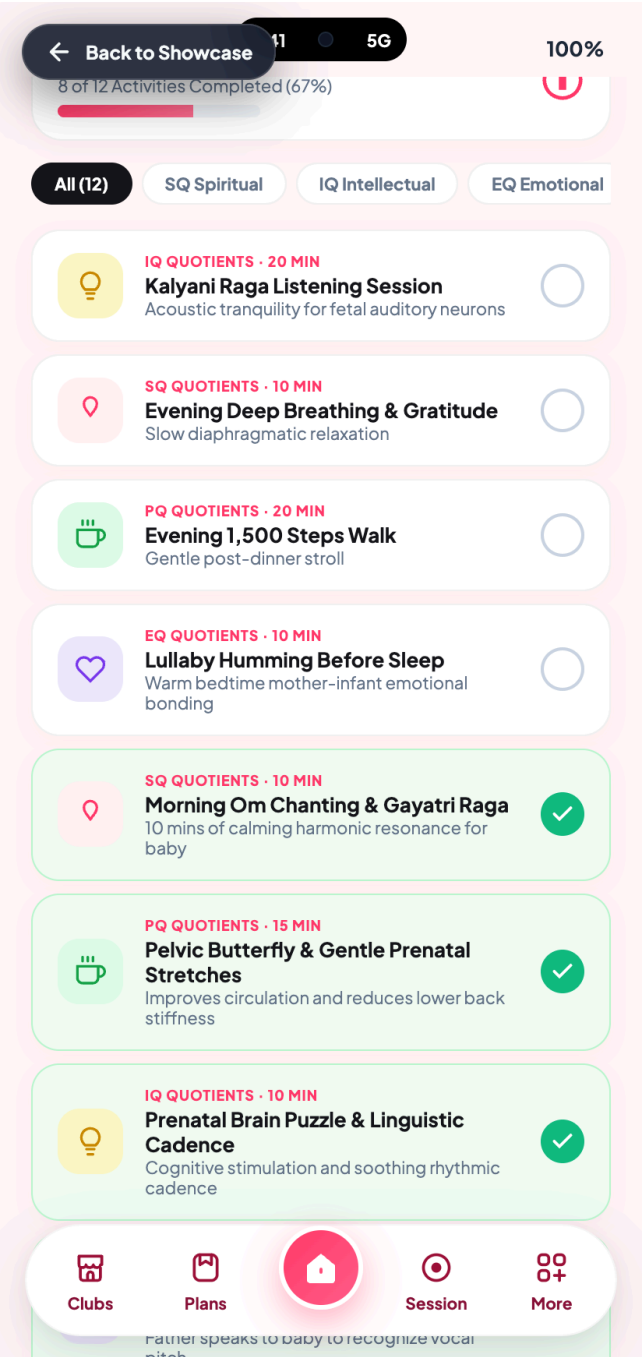
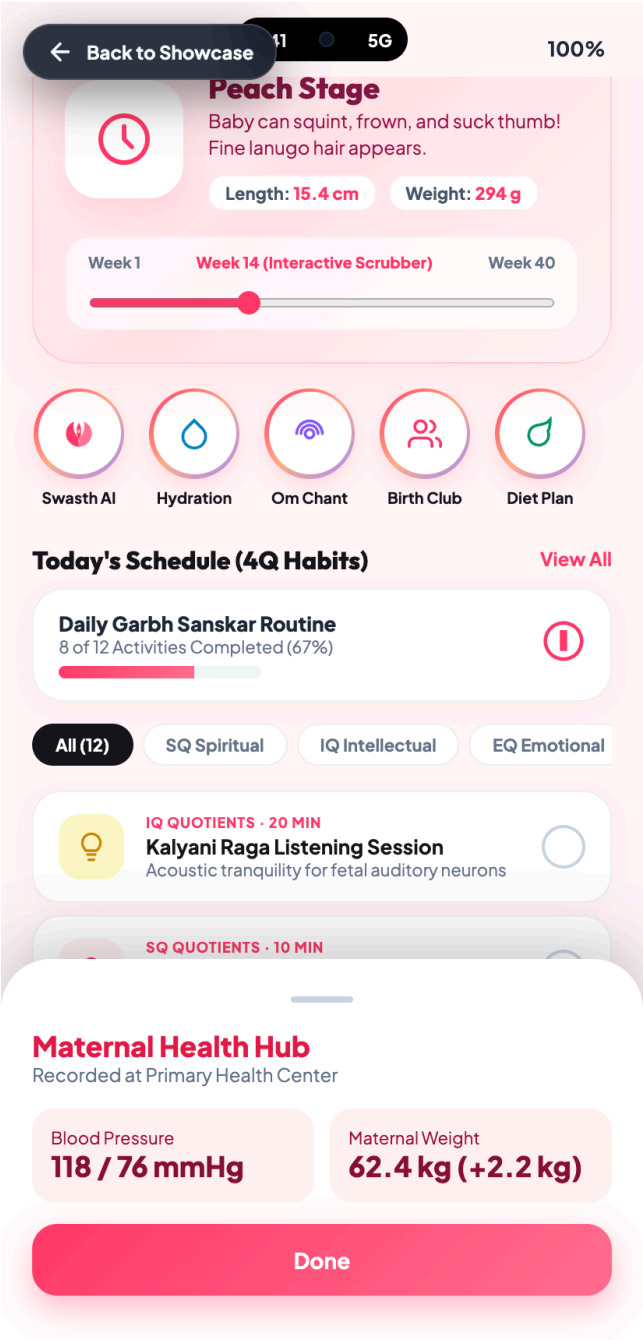
SWASTH (Maternal Health)

The Daily Cultural Companion for Mothers & Caregivers: 40-Week Growth, 4Q Garbh Sanskar Routines & Long-Term Care Continuity

- 🌸 **Maternal & Women's Health Track** · 1,000-Day Care Continuity
- 📱 **Modern Dual Architecture:** Flutter Mobile (60 FPS) + Responsive Web Simulation
- 🪷 **4Q Daily Habit Hook:** Spiritual, Physical, Intellectual & Emotional micro-rituals
- 🛡️ **Zero Clinical Diagnosis:** Strictly assistive with NVIDIA NeMo Guardrails
- 💻 **On-Device Privacy:** 100% offline-ready encrypted local SQLite storage



40-Week Interactive Timeline & Personalized Intent Setup



1. Multi-Method Due Date Setup

Supports Gestational age, LMP date, Conception date, or manual due date for instant personalization.

2. Tactile 40-Week Timeline

Interactive week scrubber with culturally relatable fruit analogies (Peach at Week 14, 8.7cm, 43g).

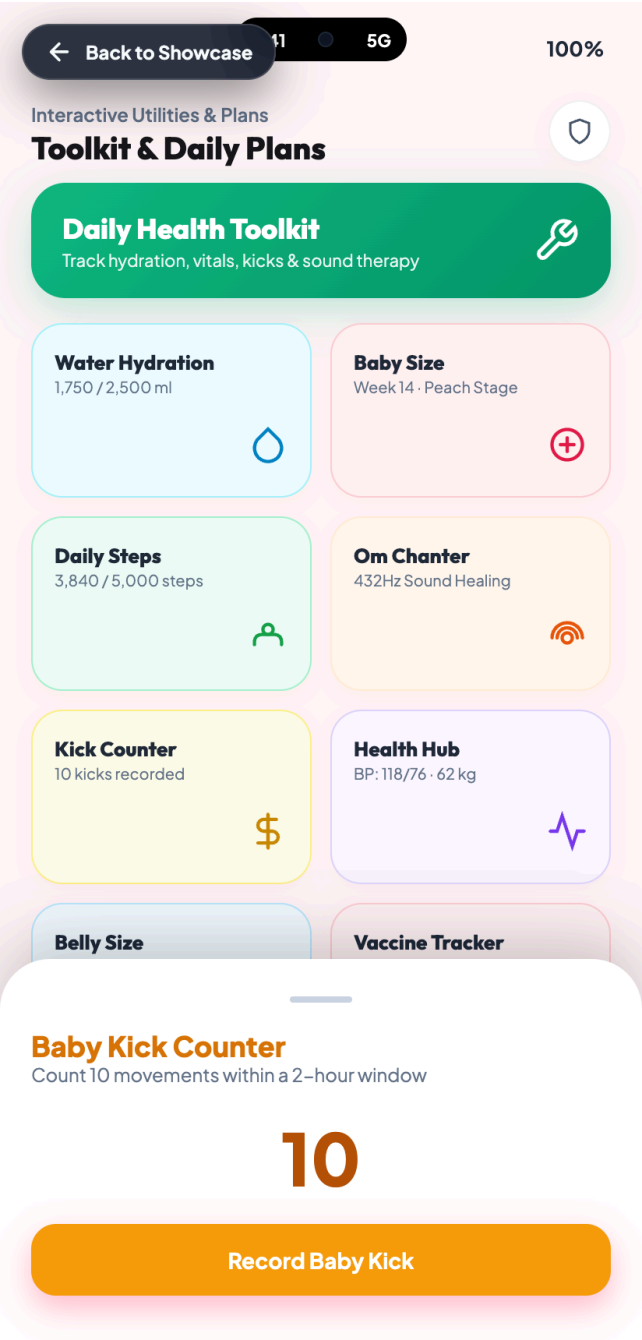
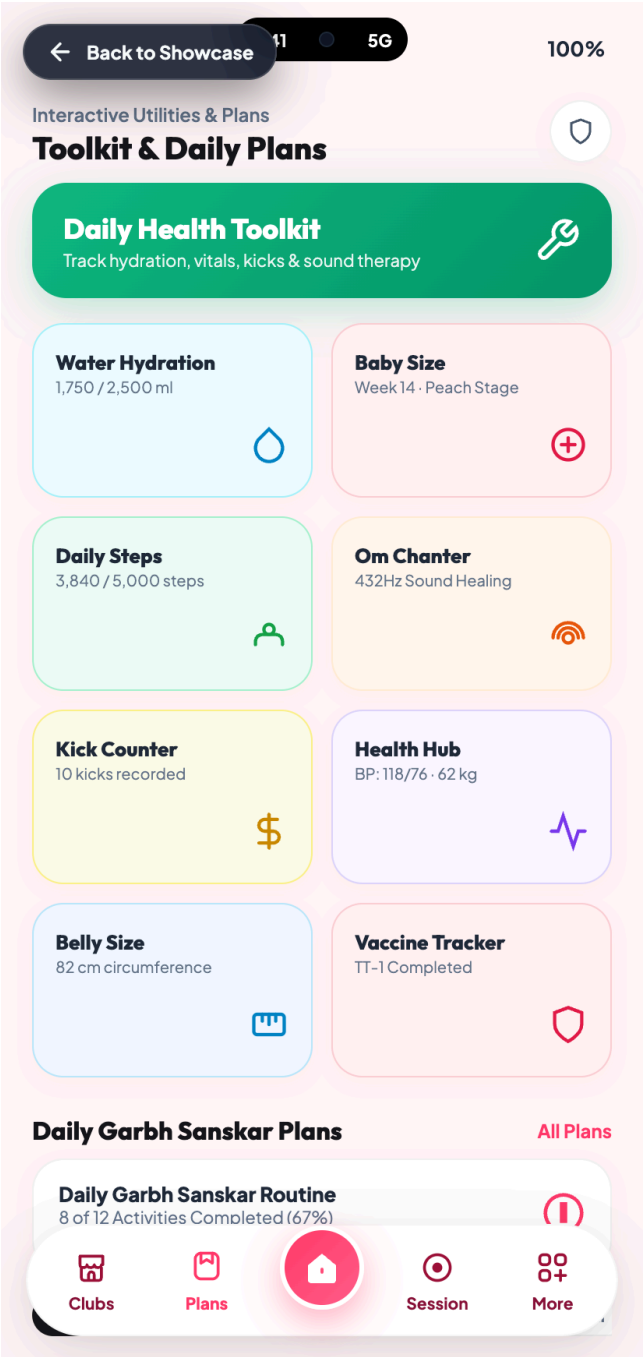
3. Gestational Countdown

Real-time Day 98 of 280 counter and upcoming primary health center checkup alerts.

4. Maternal Story Rings

One-tap vector SVG action rings for Hydration, Om Chanting, Birth Clubs, and Diet.

Interactive Health Toolkit & 4Q Garbh Sanskar Habit Engine



8-Tool Health Toolkit

Hydration 2,500ml tracker, Steps pedometer, 432Hz Om chanter, Kick counter, Health Hub BP/weight, Belly size, and TT vaccine tracker.

4Q Cultural Science Routine

12 daily habits categorized into Spiritual (SQ), Physical (PQ), Intellectual (IQ), and Emotional (EQ) quotients.

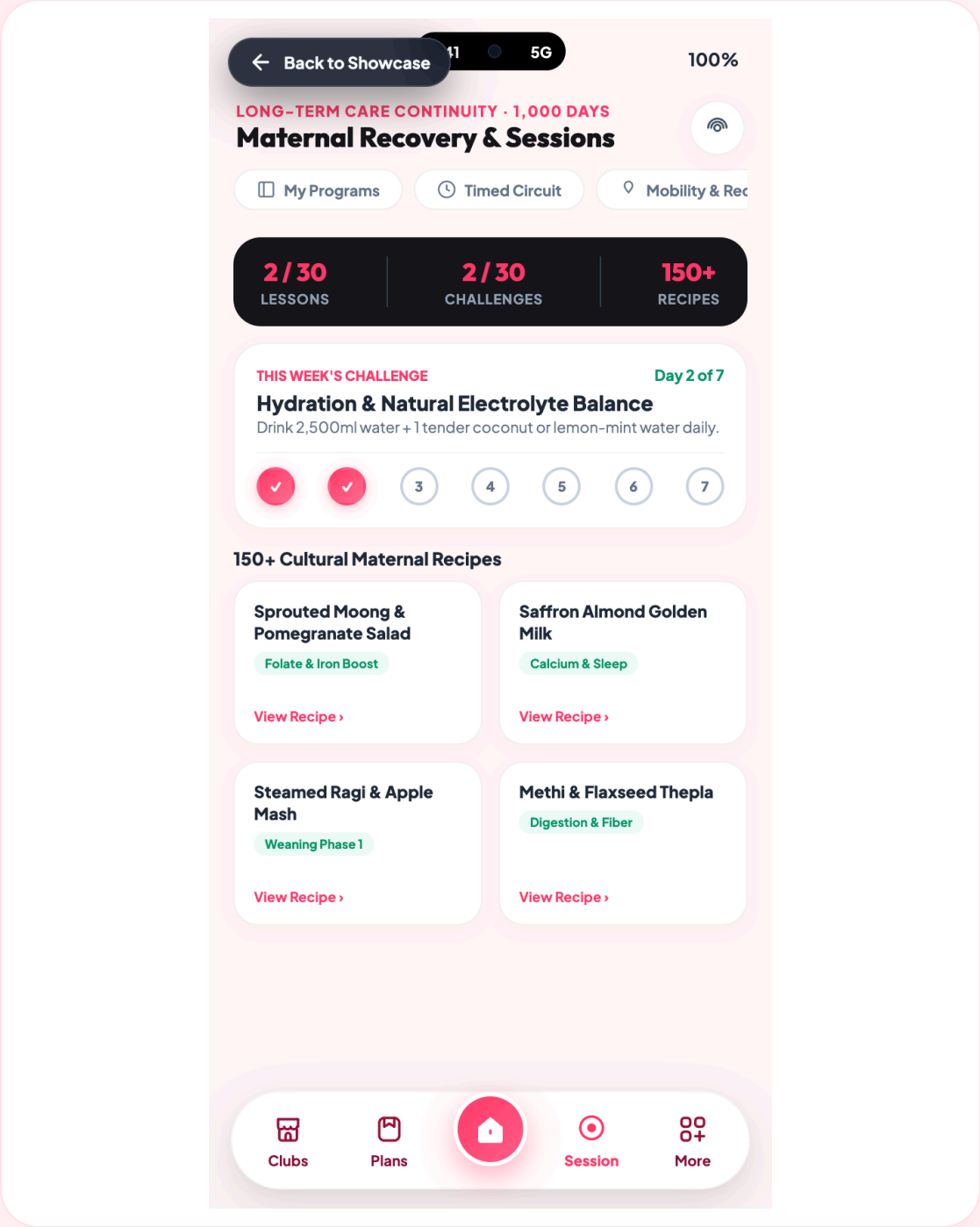
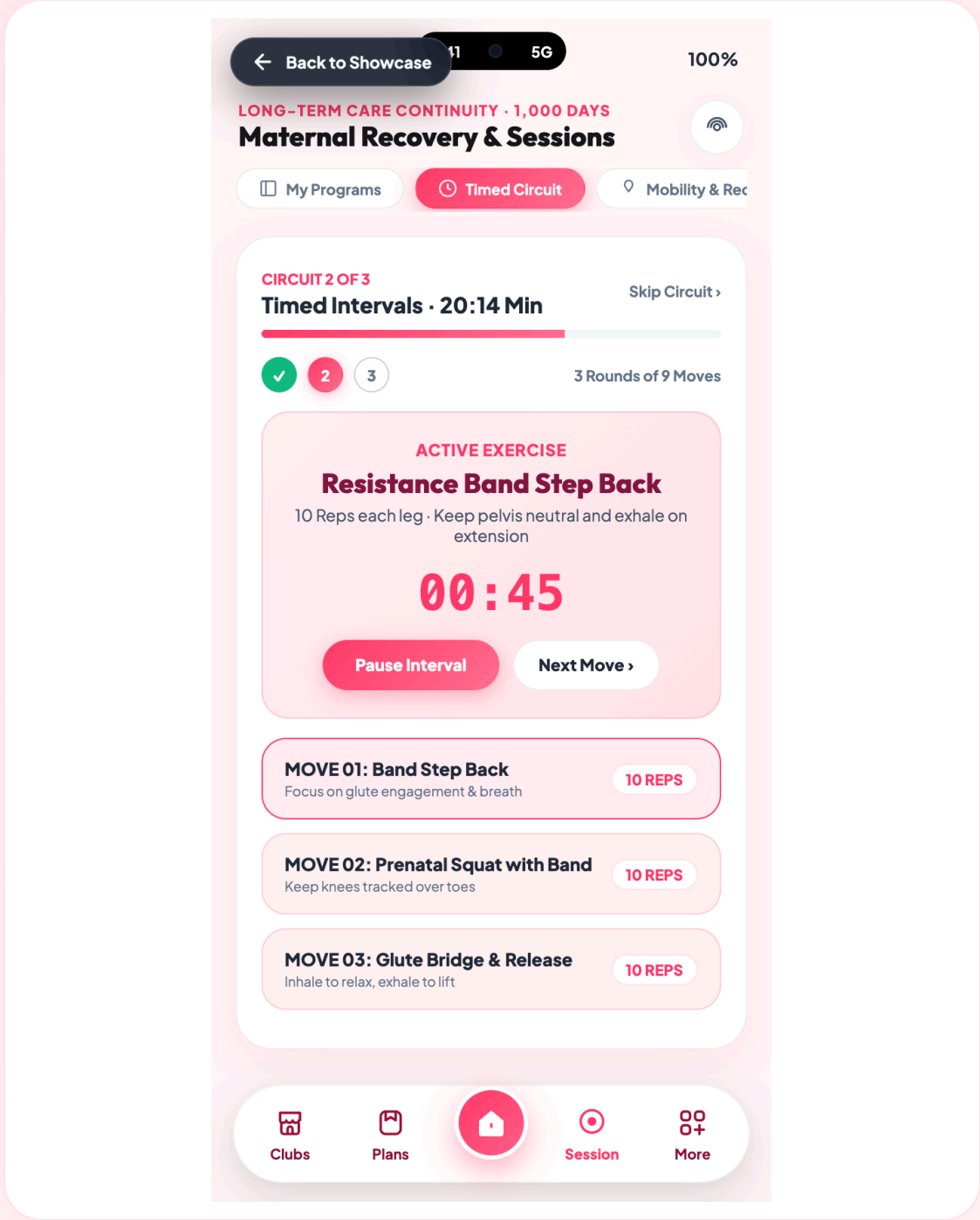
Real-Time Completion Sync

Dynamic progress card (8 of 12 completed, 67%) updating on-device SQLite database instantly.

Zero Emoji Standard

100% clean outline/duotone vector SVGs with pastel badge geometry.

Maternal Recovery Programs, Timed Circuits & 150+ Recipes



Programs for Every Stage

Beginner, Intermediate, Advanced, Pregnancy, Barre, and Diastasis-safe Core Rebuild programs.

Timed Interval Circuit Console

20:14 min guided workout player with 3 rounds of 9 moves, live countdown timer, and posture cues.

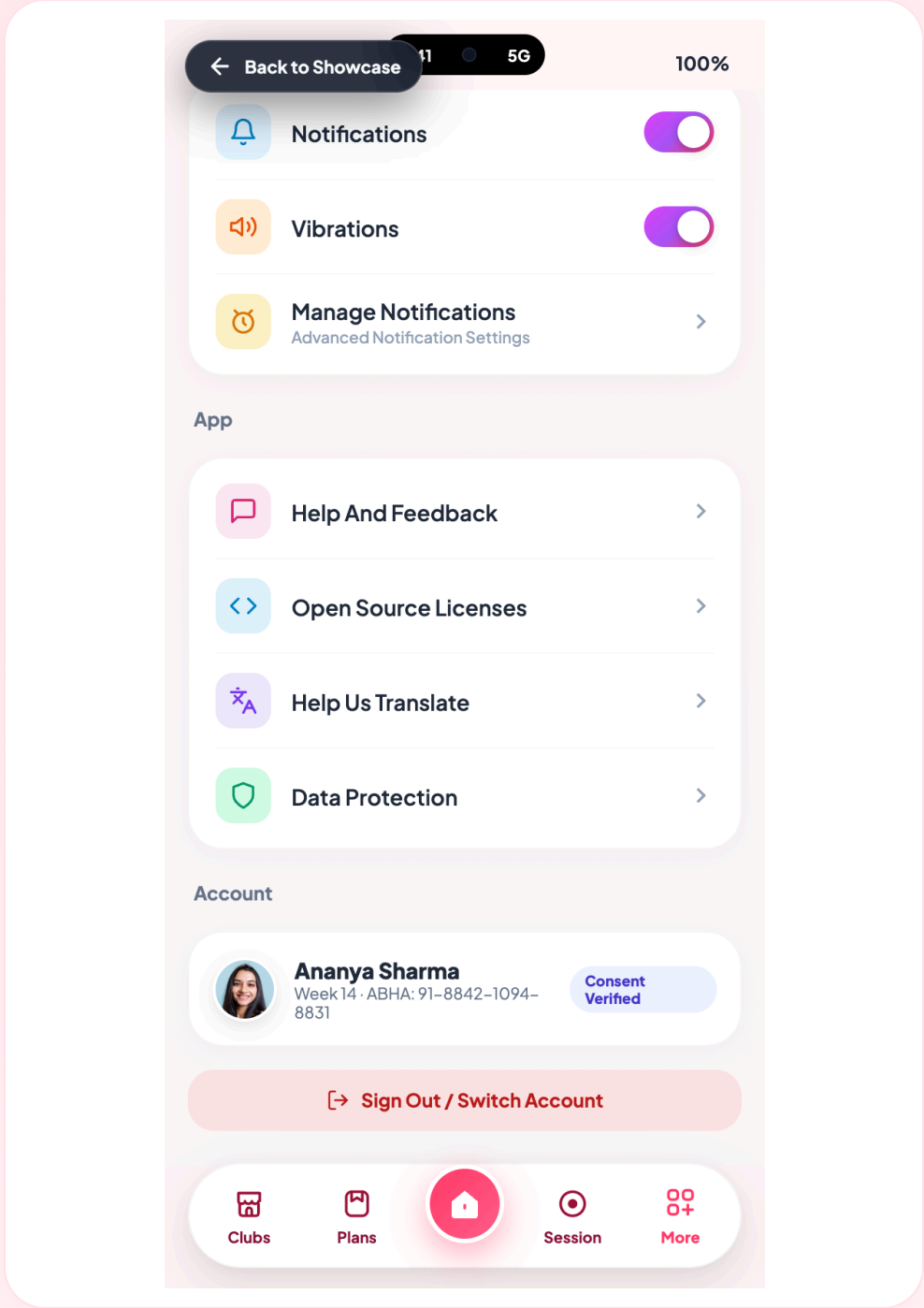
Specialist Mobility & Recovery

Pelvic floor release, prenatal yoga, and myofascial tension relief led by certified physios.

150+ Cultural Maternal Recipes

High-folate Moong salads, Kashmiri Saffron Golden Milk, and infant weaning Ragi mashes paired with Swasth AI.

Enterprise Architecture, ABHA Data Protection & 60-90 Day Impact



Open Enterprise Architecture

- **Flutter Mobile (iOS/Android):** 60 FPS fluid rendering with local SQLite persistence.
- **React/TypeScript Web:** Responsive standalone simulation portal.
- **Python (FastAPI):** Single unified backend with sub-5ms latency.
- **LangGraph + vLLM:** NVIDIA Nemotron and Llama 3 state-aware agent pipeline.
- **Indic NLP:** Whisper open voice notes & IndicTrans2 (10+ languages).
- **ABDM/ABHA Compliance:** Immutable consent-verified audit ledger.

Safety & 60-90 Day Impact

- **0% Diagnostic Advice:** Strict non-clinical scope with PHC referral routing.
- **NVIDIA NeMo Guardrails:** Intercepts severe symptoms and red flags instantly.
- **85% Drop in Administrative Burden:** Automated due lists for frontline ASHA workers.
- **90%+ Child Vaccine Adherence:** Drastically cuts the 6/10/14-week drop-off cliff.
- **Hyperlocal Birth Clubs:** Mutual peer guidance across 9,388+ online mothers.
- **100% Offline Functional:** Never blocked by low connectivity in rural centers.